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In [1]:
```
# Import necessary libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_curve, auc
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier, AdaBoostClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
import requests
import io
from imblearn.over_sampling import SMOTE
```

In [2]:
```
!gdown --fuzzy --output uber_dataset.csv https://drive.google.com/file/d/13KW_CuVVqU3e2coQkUwhD_F-w4A_gcoo/view
```

```
Downloading...
From: https://drive.google.com/uc?id=13KW_CuVVqU3e2coQkUwhD_F-w4A_gcoo
To: /content/uber_dataset.csv
100% 25.5M/25.5M [00:00<00:00, 136MB/s]
```

In [3]:
```
data = pd.read_csv('/content/uber_dataset.csv')
```

In [4]:
```
#cleaning column names
def column_name_cleaning(df):
  df.columns=df.columns.str.strip()
  df.columns=df.columns.str.lower()
  df.columns=df.columns.str.replace(' ','_')
  return df
data=column_name_cleaning(data)
```

In [5]:
```
data
```

Out[5]:

	date	time	booking_id	booking_status	customer_id	vehicle_type	pickup_location	drop_location	avg_vtat	avg_cta
0	2024-03-23	12:29:38	"CNR5884300"	No Driver Found	"CID1982111"	eBike	Palam Vihar	Jhilmil	NaN	Nal
1	2024-11-29	18:01:39	"CNR1326809"	Incomplete	"CID4604802"	Go Sedan	Shastri Nagar	Gurgaon Sector 56	4.9	14.
2	2024-08-23	08:56:10	"CNR8494506"	Completed	"CID9202816"	Auto	Khandsa	Malviya Nagar	13.4	25.
3	2024-10-21	17:17:25	"CNR8906825"	Completed	"CID2610914"	Premier Sedan	Central Secretariat	Inderlok	13.1	28.
4	2024-09-16	22:08:00	"CNR1950162"	Completed	"CID9933542"	Bike	Ghitorni Village	Khan Market	5.3	19.
...	...	...	...	...	...	...	...	...	...	.
149995	2024-11-11	19:34:01	"CNR6500631"	Completed	"CID4337371"	Go Mini	MG Road	Ghitorni	10.2	44.
149996	2024-11-24	15:55:09	"CNR2468611"	Completed	"CID2325623"	Go Mini	Golf Course Road	Akshardham	5.1	30.
149997	2024-09-18	10:55:15	"CNR6358306"	Completed	"CID9925486"	Go Sedan	Satguru Ram Singh Marg	Jor Bagh	2.7	23.
149998	2024-10-05	07:53:34	"CNR3030099"	Completed	"CID9415487"	Auto	Ghaziabad	Saidulajab	6.9	39.
149999	2024-03-10	15:38:03	"CNR3447390"	Completed	"CID4108667"	Premier Sedan	Ashok Park Main	Gurgaon Sector 29	3.5	33.

150000 rows × 21 columns

In [6]:
```
data.describe()
```

Out[6]:

	avg_vtat	avg_ctat	cancelled_rides_by_customer	cancelled_rides_by_driver	incomplete_rides	booking_value	ride_d
count	139500.000000	102000.000000	10500.0	27000.0	9000.0	102000.000000	102000
mean	8.456352	29.149636	1.0	1.0	1.0	508.295912	24
std	3.773564	8.902577	0.0	0.0	0.0	395.805774	14
min	2.000000	10.000000	1.0	1.0	1.0	50.000000	1
25%	5.300000	21.600000	1.0	1.0	1.0	234.000000	12
50%	8.300000	28.800000	1.0	1.0	1.0	414.000000	23
75%	11.300000	36.800000	1.0	1.0	1.0	689.000000	36
max	20.000000	45.000000	1.0	1.0	1.0	4277.000000	50

In [7]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150000 entries, 0 to 149999
Data columns (total 21 columns):
 #   Column                             Non-Null Count   Dtype
---  ------                             --------------   -----
 0   date                               150000 non-null  object
 1   time                               150000 non-null  object
 2   booking_id                         150000 non-null  object
 3   booking_status                     150000 non-null  object
 4   customer_id                        150000 non-null  object
 5   vehicle_type                       150000 non-null  object
 6   pickup_location                    150000 non-null  object
 7   drop_location                      150000 non-null  object
 8   avg_vtat                           139500 non-null  float64
 9   avg_ctat                           102000 non-null  float64
 10  cancelled_rides_by_customer        10500 non-null   float64
 11  reason_for_cancelling_by_customer  10500 non-null   object
 12  cancelled_rides_by_driver          27000 non-null   float64
 13  driver_cancellation_reason         27000 non-null   object
 14  incomplete_rides                   9000 non-null    float64
 15  incomplete_rides_reason            9000 non-null    object
 16  booking_value                      102000 non-null  float64
 17  ride_distance                      102000 non-null  float64
 18  driver_ratings                     93000 non-null   float64
 19  customer_rating                    93000 non-null   float64
 20  payment_method                     102000 non-null  object
dtypes: float64(9), object(12)
memory usage: 24.0+ MB
```

In [8]:
```
# checking for missing values
# data.isna().sum()
100*data.notna().sum()/len(data)
# 100*data.isna().sum()/len(data)
```

Out[8]:

	0
date	100.0
time	100.0
booking_id	100.0
booking_status	100.0
customer_id	100.0
vehicle_type	100.0
pickup_location	100.0
drop_location	100.0
avg_vtat	93.0
avg_ctat	68.0
cancelled_rides_by_customer	7.0
reason_for_cancelling_by_customer	7.0
cancelled_rides_by_driver	18.0
driver_cancellation_reason	18.0
incomplete_rides	6.0
incomplete_rides_reason	6.0
booking_value	68.0
ride_distance	68.0
driver_ratings	62.0
customer_rating	62.0
payment_method	68.0

dtype: float64

From preliminary data observation, we can understand that:

7% rides are cancelled by customers

18% rides are cancelled by drivers

6% are incomplete rides with reasons ['Vehicle Breakdown', 'Other Issue', 'Customer Demand']

In [9]:

```
data[data['ride_distance'].isna()]['booking_status'].unique()
data[data['booking_status']=='Completed']['ride_distance'].isna().unique()
```

Out[9]:

```
array([False])
```

In [10]:

```
data[data['booking_value'].isna()]['booking_status'].unique()
data[data['booking_status']=='Completed']['booking_value'].isna().unique()
```

Out[10]:

```
array([False])
```

In [11]:

```
data[data['payment_method'].isna()]['booking_status'].unique()
data[data['booking_status']=='Completed']['payment_method'].isna().unique()
```

Out[11]:

```
array([False])
```

In [12]:

```
data[data['driver_ratings'].isna()]['booking_status'].unique()
data[data['booking_status']=='Completed']['driver_ratings'].isna().unique()
```

Out[12]:

```
array([False])
```

In [13]:

```
data[data['customer_rating'].isna()]['booking_status'].unique()
# data[data['booking_status']=='Completed']['customer_rating'].isna().unique()
```

Out[13]:

```
array(['No Driver Found', 'Incomplete', 'Cancelled by Driver',
       'Cancelled by Customer'], dtype=object)
```

In [14]:
```
data[data['booking_status']=='No Driver Found']['avg_vtat'].unique()
```
Out[14]:
```
array([nan])
```
In [15]:
```
data[data['booking_status']=='Incomplete']['avg_vtat'].isna().unique()
```
Out[15]:
```
array([False])
```
In [16]:
```
data[data['booking_status']=='Cancelled by Driver']['avg_vtat'].isna().unique()
```
Out[16]:
```
array([False])
```
In [17]:
```
data[data['booking_status']=='Cancelled by Customer']['avg_vtat'].isna().unique()
```
Out[17]:
```
array([False])
```
In [18]:
```
data[data['avg_ctat'].isna()]['booking_status'].unique()
data[data['booking_status']=='Completed']['avg_ctat'].isna().unique()
```
Out[18]:
```
array([False])
```
In [19]:
```
data[data['booking_status']=='No Driver Found']['avg_ctat'].isna().unique()
```
Out[19]:
```
array([ True])
```
In [20]:
```
data[data['booking_status']=='Cancelled by Driver']['avg_ctat'].isna().unique()
```
Out[20]:
```
array([ True])
```
In [21]:
```
data[data['booking_status']=='Cancelled by Customer']['avg_ctat'].isna().unique()
```
Out[21]:
```
array([ True])
```

32% data in columns: ['ride_distance', 'booking_value', 'payment_method'] is null. These are rows that have 'booking_status' other than "Completed".

38% data in columns: ['driver_ratings', 'customer_rating'] is null. These are rows with 'booking_status' other than "Completed".

avg_vtat has 7% nulls, all for rows with 'booking_status'='No Driver Found', and for other 'booking_status' it is not null. This tells us that there is no discrapency in vtat data.

avg_ctat has 32% nulls, all for rows with 'booking_status'="Completed", and for others it is not null. Indicating no discrapency in avg_ctat values.

No discrepancy in 'ride_distance' column, it is not null for rows with booking status "Completed" and null for others.

Similarly, for 'booking_value', 'payment_method', 'driver_ratings', 'customer_rating' columns.

In [22]:
```
# Customer cancellation rate creation
data['is_customer_cancel']=data['booking_status'].apply(lambda x: 1 if x=='Cancelled by Customer' else 0)
customer_cancel_group=data.groupby('customer_id').agg(cancel_count=('is_customer_cancel','sum'))

customer_book_group=data.groupby('customer_id').agg(book_count=('booking_id','count'))

data['cancel_count']=data['customer_id'].map(customer_cancel_group['cancel_count'])
data['book_count']=data['customer_id'].map(customer_book_group['book_count'])
data['customer_cancel_rate']=(100*data['cancel_count']/data['book_count']).round(2)
# data['customer_cancel_rate'].unique()

data.drop(columns=['is_customer_cancel','cancel_count','book_count'], axis=1, inplace=True)
```
In [23]:
```
# Binarizing booking status
data['booking_status_bin']=data['booking_status'].apply(lambda x: 1 if x=='Completed' else 0)
```
In [24]:
```
potential_leakgae_cols=['driver_ratings', 'customer_rating', 'booking_value', 'payment_method', 'ride_distance']  # the
```

Doubt: Should we delete these leakgae columsn or not?

Data Handling

In [25]:
```
data.head()
```

Out[25]:

	date	time	booking_id	booking_status	customer_id	vehicle_type	pickup_location	drop_location	avg_vtat	avg_ctat	...
0	2024-03-23	12:29:38	"CNR5884300"	No Driver Found	"CID1982111"	eBike	Palam Vihar	Jhilmil	NaN	NaN	...
1	2024-11-29	18:01:39	"CNR1326809"	Incomplete	"CID4604802"	Go Sedan	Shastri Nagar	Gurgaon Sector 56	4.9	14.0	...
2	2024-08-23	08:56:10	"CNR8494506"	Completed	"CID9202816"	Auto	Khandsa	Malviya Nagar	13.4	25.8	...
3	2024-10-21	17:17:25	"CNR8906825"	Completed	"CID2610914"	Premier Sedan	Central Secretariat	Inderlok	13.1	28.5	...
4	2024-09-16	22:08:00	"CNR1950162"	Completed	"CID9933542"	Bike	Ghitorni Village	Khan Market	5.3	19.6	...

5 rows × 23 columns

In [26]:
```
data.info()
```

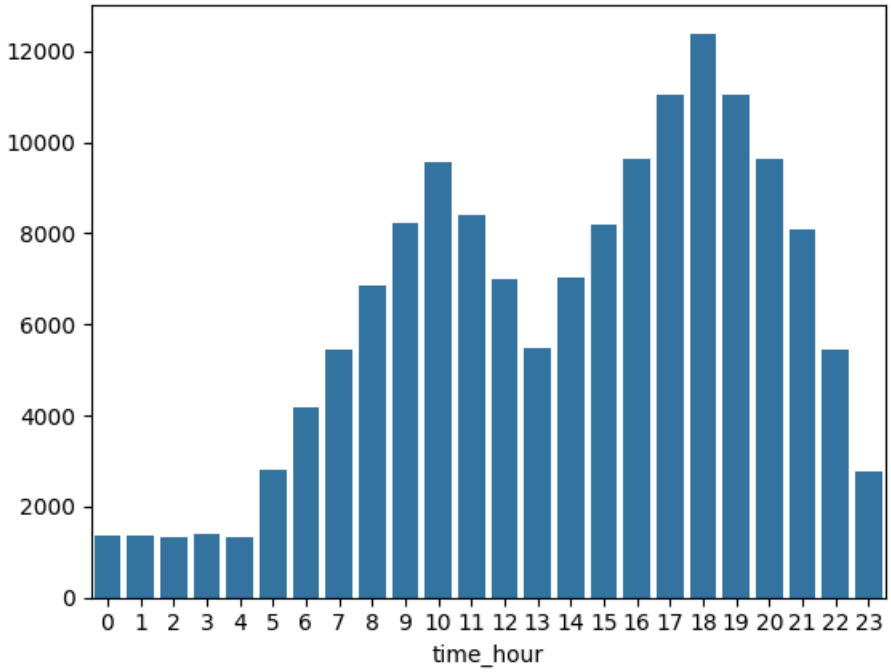
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150000 entries, 0 to 149999
Data columns (total 23 columns):
 #   Column                             Non-Null Count   Dtype
---  ------                             --------------   -----
 0   date                               150000 non-null  object
 1   time                               150000 non-null  object
 2   booking_id                         150000 non-null  object
 3   booking_status                     150000 non-null  object
 4   customer_id                        150000 non-null  object
 5   vehicle_type                       150000 non-null  object
 6   pickup_location                    150000 non-null  object
 7   drop_location                      150000 non-null  object
 8   avg_vtat                           139500 non-null  float64
 9   avg_ctat                           102000 non-null  float64
 10  cancelled_rides_by_customer        10500 non-null   float64
 11  reason_for_cancelling_by_customer  10500 non-null   object
 12  cancelled_rides_by_driver          27000 non-null   float64
 13  driver_cancellation_reason         27000 non-null   object
 14  incomplete_rides                   9000 non-null    float64
 15  incomplete_rides_reason            9000 non-null    object
 16  booking_value                      102000 non-null  float64
 17  ride_distance                      102000 non-null  float64
 18  driver_ratings                     93000 non-null   float64
 19  customer_rating                    93000 non-null   float64
 20  payment_method                     102000 non-null  object
 21  customer_cancel_rate               150000 non-null  float64
 22  booking_status_bin                 150000 non-null  int64
dtypes: float64(10), int64(1), object(12)
memory usage: 26.3+ MB
```

In [27]:
```
data['date']=pd.to_datetime(data['date'])
```

In [28]:
```
data['time_hour']=pd.to_datetime(data['time']).dt.hour
```

```
/tmp/ipython-input-1265285900.py:1: UserWarning: Could not infer format, so each element will be parsed individually, fall
ing back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.
  data['time_hour']=pd.to_datetime(data['time']).dt.hour
```

In [29]:
```
hour_demand=data.groupby('time_hour').size()
sns.barplot(x=hour_demand.index, y=hour_demand.values)
plt.show()
```



In [30]:

```
time_labels=['Late Night','Morning Peak', 'Midday', 'Evening Rush','Night']
hour_bins=[0,5,11,15,18,23]
data['time_slot_of_day']=pd.cut(data['time_hour'],bins=hour_bins, labels=time_labels, right=True, include_lowest=True)
```

In [31]:

```
# pickup location density score creation
# data['pickup_location'].nunique()#176 pickup locations
# pd.Series(data['pickup_location'].unique()).sort_values()
pickup_demand=data.groupby('pickup_location').agg(demand=('booking_id','count'), completed=('booking_status_bin','sum')
pickup_demand['completion_rate']=100*pickup_demand['completed']/pickup_demand['demand']
pickup_demand.sort_values(by='completion_rate',ascending=False)
#capturing only driver demand and completion rate, not focussing on driver pressures
# completion rate is not a great signal for defining location density
pickup_demand['pickup_location_density_score']=np.log1p(pickup_demand['demand'])*pickup_demand['completion_rate']
pickup_demand.sort_values(by='pickup_location_density_score',ascending=False)
#ensures high demand and high completion rate to score better
data['pickup_location_density_score']=data['pickup_location'].map(pickup_demand['pickup_location_density_score'])
data['pickup_location_density_score']
```

Out[31]:

	pickup_location_density_score
0	429.711010
1	426.467361
2	433.496016
3	418.129009
4	418.981746
...	...
149995	412.451664
149996	416.590063
149997	414.421293
149998	413.894262
149999	435.997034

150000 rows × 1 columns

dtype: float64

In [32]:

```
# drop location density score creation
# data['drop_location'].nunique()#176 pickup locations
# pd.Series(data['drop_location'].unique()).sort_values()
drop_demand=data.groupby('drop_location').agg(demand=('booking_id','count'), completed=('booking_status_bin','sum'))
drop_demand['completion_rate']=100*drop_demand['completed']/drop_demand['demand']
drop_demand.sort_values(by='completion_rate',ascending=False)
```

```
#capturing only driver demand and completion rate, not focussing on driver pressures
# completion rate is not a great signal for defining location density
drop_demand['drop_location_density_score']=np.log1p(drop_demand['demand'])*drop_demand['completion_rate']
drop_demand.sort_values(by='drop_location_density_score',ascending=False)

# #ensures high demand and high completion rate to score better
data['drop_location_density_score']=data['drop_location'].map(drop_demand['drop_location_density_score'])
data['drop_location_density_score']
```

Out[32]:

	drop_location_density_score
0	425.669442
1	415.108026
2	424.041694
3	420.640016
4	430.170968
...	...
149995	412.686805
149996	436.291024
149997	435.309489
149998	420.010790
149999	389.166542

150000 rows × 1 columns

dtype: float64

pickup and drop locations are encoded using historical density scores that combine booking volume and completion rate per zone. This approach captures both geographical activity density and behaviorial similarity, serving same purpose as target encoding.

In [33]:

```
data['time_slot_of_day'].unique()
```

Out[33]:

```
['Midday', 'Evening Rush', 'Morning Peak', 'Night', 'Late Night']
Categories (5, object): ['Late Night' < 'Morning Peak' < 'Midday' < 'Evening Rush' < 'Night']
```

In [34]:

```
# trip duration feature definition: defined with estimations on vehicle speed and speed multiplier according to time
# data['ride_distance'].unique()
vehicle_base_speed = {
    'eBike': 35,
    'Bike': 32,
    'Go Mini': 28,
    'Go Sedan': 27,
    'Premier Sedan': 27,
    'Auto': 22,
    'Uber XL': 24
}
time_speed_multiplier = {
    'Late Night': 1.3,
    'Morning Peak': 0.6,
    'Midday': 1.0,
    'Evening Rush': 0.6,
    'Night': 1.2
}
data['base_speed'] = data['vehicle_type'].map(vehicle_base_speed)
data['traffic_multiplier'] = data['time_slot_of_day'].map(time_speed_multiplier)

data['expected_speed'] = (
    data['base_speed'] * data['traffic_multiplier']
)
data['trip_duration'] = 60*data['ride_distance'] / data['expected_speed']
data.drop(columns=['base_speed', 'traffic_multiplier', 'expected_speed'], axis=1, inplace=True)
data['trip_duration']
```

Out[34]:

	trip_duration
0	NaN
1	21.222222
2	61.727273
3	126.000000
4	75.328125
...	...
149995	71.571429
149996	45.664286
149997	59.000000
149998	207.000000
149999	47.088889

150000 rows × 1 columns

dtype: float64

In [35]:

```
#day type feature
data['day_type']=(data['date'].dt.day_of_week).apply(lambda x: 'Weekday' if x<5 else 'Weekend')
```

Data Cleaning

In [36]:

```
potential_leakgae_cols
```

Out[36]:

```
['driver_ratings',
 'customer_rating',
 'booking_value',
 'payment_method',
 'ride_distance']
```

In [37]:

```
# binarize booking_value column
data['is_booking_value_available']=data['booking_value'].apply(lambda x: 1 if pd.notnull(x) else 0)
data['booking_value']=data['booking_value'].apply(lambda x: -1 if pd.isnull(x) else x) # -1 means missing value
data[data['booking_value']==-1][['booking_value','is_booking_value_available']]
```

Out[37]:

	booking_value	is_booking_value_available
0	-1.0	0
8	-1.0	0
11	-1.0	0
12	-1.0	0
17	-1.0	0
...	...	...
149979	-1.0	0
149980	-1.0	0
149982	-1.0	0
149990	-1.0	0
149992	-1.0	0

48000 rows × 2 columns

In [38]:

```
#binarize ride distance column
data['is_ride_distance_available']=data['ride_distance'].apply(lambda x: 1 if pd.notnull(x) else 0)
data['ride_distance']=data['ride_distance'].apply(lambda x: -1 if pd.isnull(x) else x) # -1 means missing value
data[data['ride_distance']==-1][['ride_distance','is_ride_distance_available']]
```

Out[38]:

	ride_distance	is_ride_distance_available
0	-1.0	0
8	-1.0	0
11	-1.0	0
12	-1.0	0
17	-1.0	0
...	...	...
149979	-1.0	0
149980	-1.0	0
149982	-1.0	0
149990	-1.0	0
149992	-1.0	0

48000 rows × 2 columns

In [39]:

```
# imputing avg_vtat with mean
data['avg_vtat']=data.groupby('vehicle_type')['avg_vtat'].transform(lambda x: x.fillna(x.median()))
```

In [40]:

```
data['avg_ctat']=data.groupby('vehicle_type')['avg_ctat'].transform(lambda x: x.fillna(x.median()))
```

In [41]:

```
# data[['time','date']]
data['date_time_combined']=pd.to_datetime(data['date'].astype(str)+' '+data['time'].astype(str))
data.drop(columns=['date','time'], axis=1, inplace=True, errors='ignore')
```

In [42]:

```
# potential_leakgae_cols
data.drop(columns=['driver_ratings', 'customer_rating'], axis=1, inplace=True)
potential_leakgae_cols.remove('driver_ratings')
potential_leakgae_cols.remove('customer_rating')
```

Class balance Check

In [43]:

```
# function to split columns as cat and numerical columns
def find_cat_num_cols(df):
  # cat_cols=list(df.select_dtypes(include="object").columns)
  # num_cols=list(df.select_dtypes(exclude="object").columns)
  cat_cols=[]
  num_cols=[]
  for x in df.columns:
    if df[x].nunique()<10:
      cat_cols.append(x)
    else:
      num_cols.append(x)
  return cat_cols, num_cols
```

In [44]:

```
potential_leakgae_cols
```

Out[44]:

```
['booking_value', 'payment_method', 'ride_distance']
```

In [45]:

```
# selecting only required columns
req_cols=['booking_status', 'booking_status_bin', 'vehicle_type', 'pickup_location', 'drop_location', 'avg_vtat','avg_c
# removing leakage columns: 'cancelled_rides_by_customer', 'cancelled_rides_by_driver', 'incomplete_rides', 'booking_va
```

In [46]:

```
dataset=data[req_cols].copy()
dataset.columns
```

Out[46]:

```
Index(['booking_status', 'booking_status_bin', 'vehicle_type',
       'pickup_location', 'drop_location', 'avg_vtat', 'avg_ctat', 'time_hour',
       'time_slot_of_day', 'day_type', 'pickup_location_density_score',
       'drop_location_density_score'],
      dtype='object')
```

In [47]:

```
cat_cols, num_cols =find_cat_num_cols(dataset)
```

In [48]:
```
num_cols.remove('pickup_location')
num_cols.remove('drop_location')
```

Remove NaN from cols before finding imbalances

In [49]:
```
cat_cols
```

Out[49]:
```
['booking_status',
 'booking_status_bin',
 'vehicle_type',
 'time_slot_of_day',
 'day_type']
```

In [50]:
```
# #replacing nulls in 'cancelled_rides_by_customer','cancelled_rides_by_driver','incomplete_rides' with 0
# dataset['cancelled_rides_by_customer']=dataset['cancelled_rides_by_customer'].fillna(0)
# dataset['cancelled_rides_by_driver']=dataset['cancelled_rides_by_driver'].fillna(0)
# dataset['incomplete_rides']=dataset['incomplete_rides'].fillna(0)
```

In [51]:
```
# replacing nulls in trip_duration with 0
# dataset['trip_duration']=dataset['trip_duration'].fillna(0)
```

In [52]:
```
dataset.isna().sum()
```

Out[52]:

	0
booking_status	0
booking_status_bin	0
vehicle_type	0
pickup_location	0
drop_location	0
avg_vtat	0
avg_ctat	0
time_hour	0
time_slot_of_day	0
day_type	0
pickup_location_density_score	0
drop_location_density_score	0

dtype: int64

In [53]:
```
# dataset[dataset['payment_method'].isna()]['booking_status'].unique()
# dataset[dataset['payment_method'].isna() & dataset['booking_status']=='Cancelled by Driver']
# dataset[dataset['booking_status']=='Incomplete'][['payment_method','booking_value']]
```

Though booking_status is 'Incomplete', there are no null in 'ride_distance', 'booking_value', and 'payment_method'

In [54]:
```
dataset.drop(columns=['booking_status'], axis=1, inplace=True, errors='ignore')
```

In [55]:
```
#converting cat columns to numerical
def cat_to_num_convert(df, col_name, mapping_tuple, null_replacer_value):
  df[col_name]=df[col_name].replace(mapping_tuple).fillna(null_replacer_value).astype(int)
  return df
```

In [56]:
```
def vis_cat_cols(df, cat_cols,fig_size=(15,15), plot_col_num=3):
  plot_num_cols = plot_col_num
  plot_num_rows = len(cat_cols)//plot_num_cols+(1 if len(cat_cols)%plot_num_cols>0 else 0)
  fig, axes = plt.subplots(plot_num_rows, plot_num_cols, figsize=fig_size)
  axes = axes.flatten()
```

```
  for i, column in enumerate(cat_cols):
    sns.countplot(data=df,x=column,ax=axes[i])
    axes[i].set_title(f'Count Plot of {column}')
    axes[i].set_xlabel(column)
    axes[i].set_ylabel('Count')
    axes[i].tick_params(axis='x',rotation=90)
  for j in range(i + 1, len(axes)): # Turn off any unused subplots
    fig.delaxes(axes[j])
  plt.tight_layout()
  plt.show()
  return
```

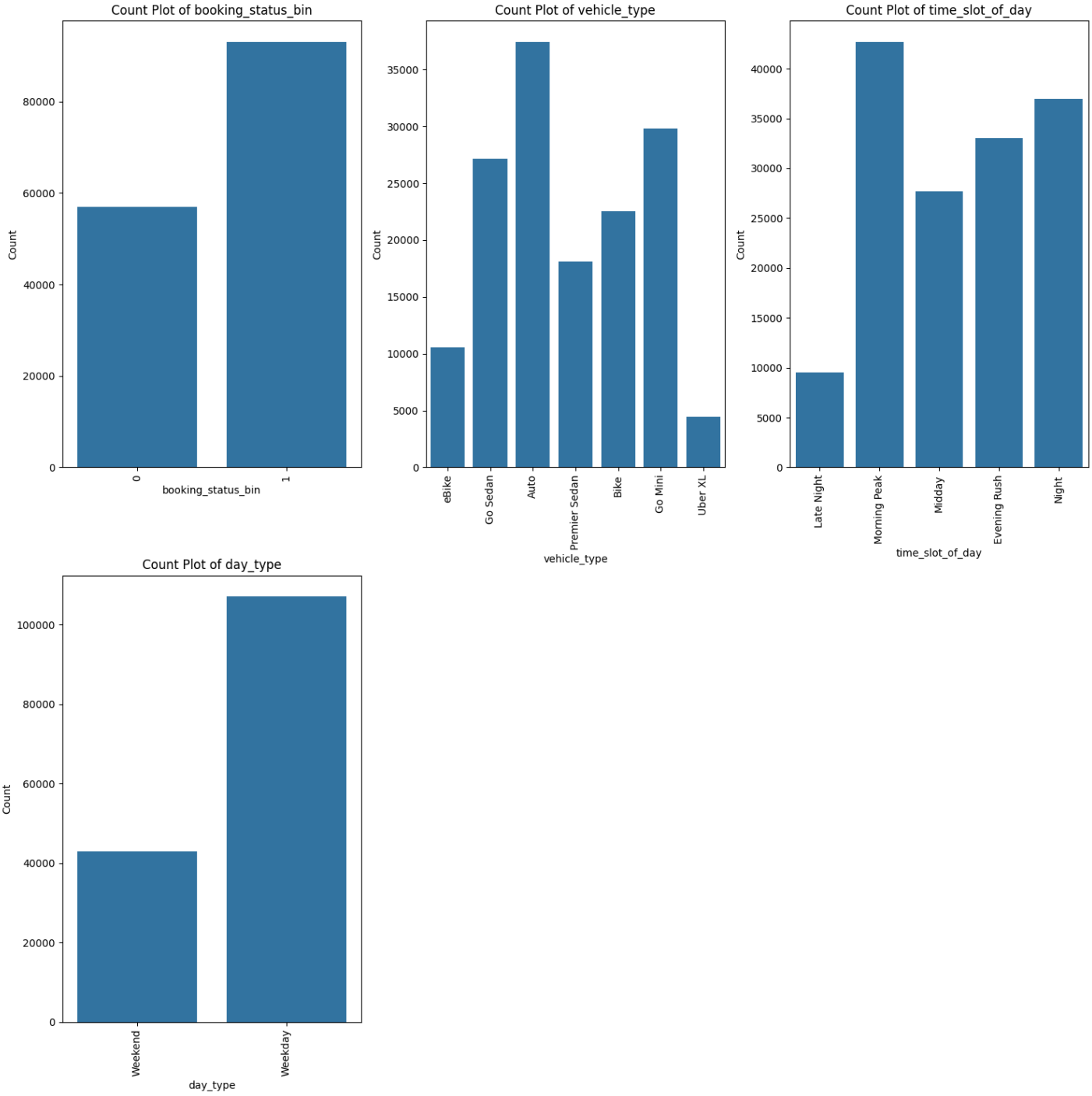
In [57]:
```
cat_cols, num_cols =find_cat_num_cols(dataset)
```

In [58]:
```
cat_cols
```

Out[58]:
```
['booking_status_bin', 'vehicle_type', 'time_slot_of_day', 'day_type']
```

In [59]:
```
cat_cols=['booking_status_bin','vehicle_type','time_slot_of_day','day_type']
```

In [60]:
```
vis_cat_cols(dataset,cat_cols)
```



In [61]:
```
def vis_num_cols(df, num_cols, bins=20,fig_size=(15,15)):
  plot_num_cols = 2
```

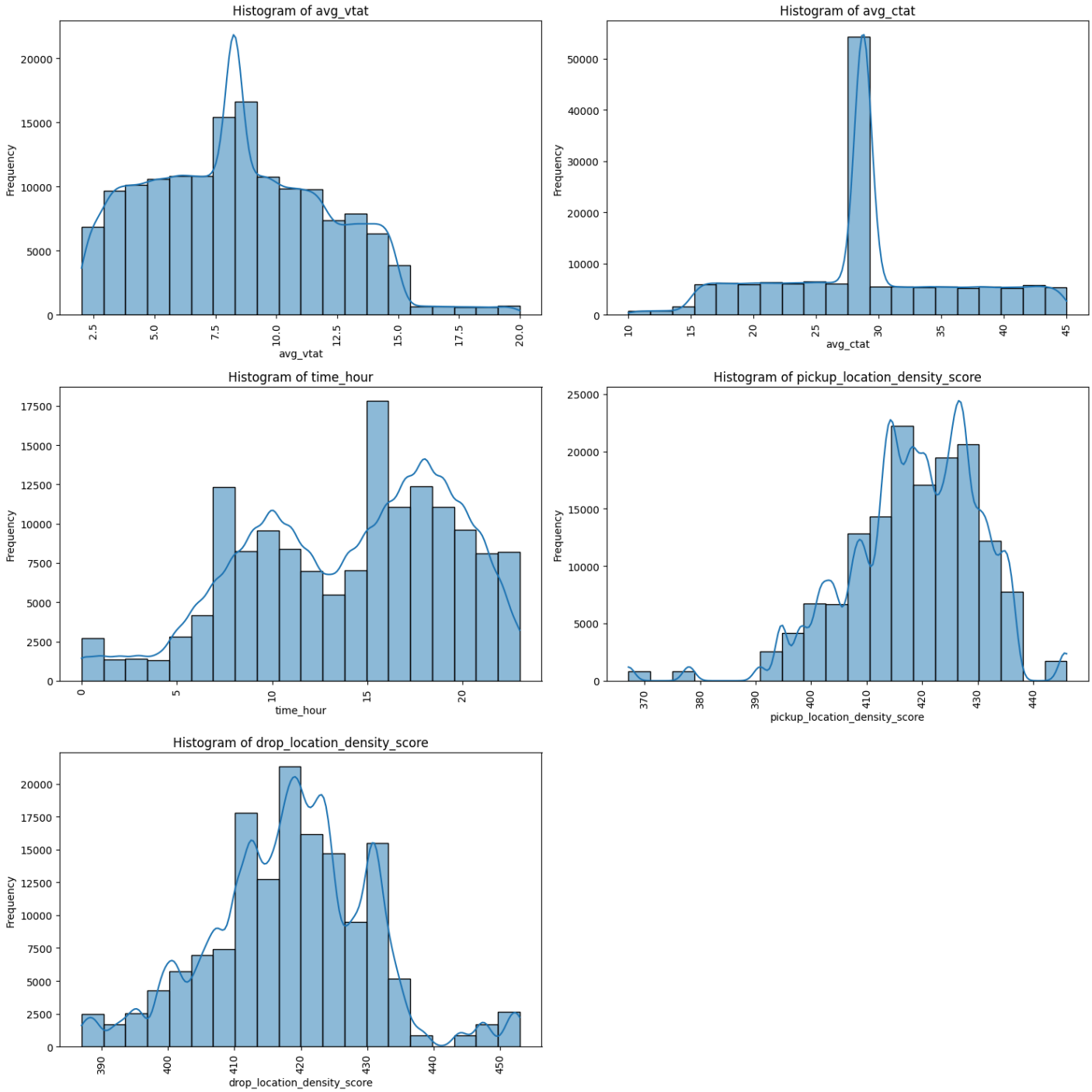
```
  plot_num_rows = len(num_cols)//plot_num_cols+(1 if len(num_cols)%plot_num_cols>0 else 0)
  fig, axes = plt.subplots(plot_num_rows, plot_num_cols, figsize=fig_size)
  axes = axes.flatten()

  for i, column in enumerate(num_cols):
    sns.histplot(data=df, x=column, bins=bins,kde=True, ax=axes[i])
    axes[i].set_title(f'Histogram of {column}')
    axes[i].set_xlabel(column)
    axes[i].set_ylabel('Frequency')
    axes[i].tick_params(axis='x',rotation=90)

  for j in range(i + 1, len(axes)): # Turn off any unused subplots
    fig.delaxes(axes[j])

  plt.tight_layout()
  plt.show()
  return
```

In [62]:

```
try:
  num_cols.remove('pickup_location')
  num_cols.remove('drop_location')
except:
  pass
```

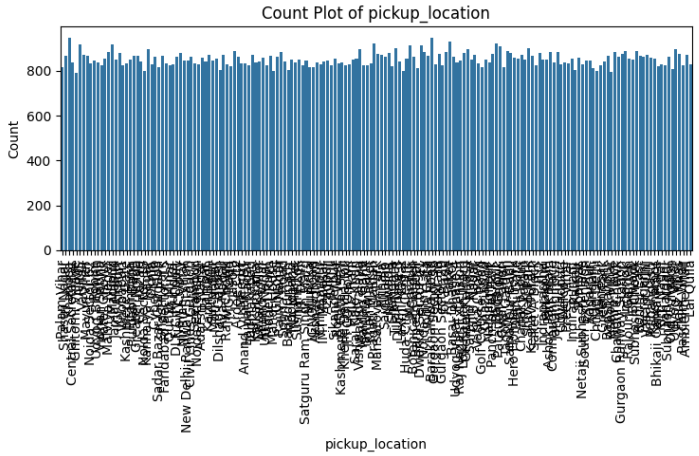
In [63]:

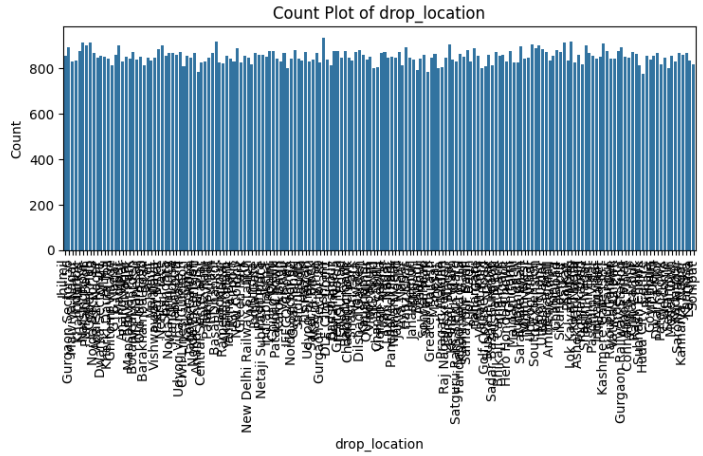
```
vis_num_cols(dataset,num_cols)
```



In [64]:

```
vis_cat_cols(dataset,['pickup_location','drop_location'],fig_size=(15,5),plot_col_num=2)
```





pickup and drop location data is roughly balanced

In [65]:
```
dataset['booking_status_bin'].value_counts()*100/len(dataset)
```

Out[65]:

booking_status_bin	count
1	62.0
0	38.0

dtype: float64

The target class is midly imbalanced, and doesn't require SMOTE automatically. I'll find recall, precision, f1_score, roc_auc and then decide on SMOTE based on underperformance on minority class.

Q1. **Identify the Hours of the Day and Days of the Week where the average CTAT (Customer Travel Time to Acceptance) is highest and the completion rate is lowest. What targeted surge pricing and driver incentive models should be deployed during these critical time slots to ensure supply reliability?**

In [66]:
```
dataset['is_weekend']=dataset['day_type'].apply(lambda x: 1 if x=='Weekend' else 0)
```

In [67]:
```
summary_week_hour=dataset.groupby(['is_weekend','time_hour']).agg(mean_avg_ctat=('avg_ctat','mean'),mean_completion_rat
summary_week_hour.head()
```

Out[67]:

is_weekend	time_hour	mean_avg_ctat	mean_completion_rate	demand
	0	28.950409	0.619632	978
	1	28.839297	0.617373	967
0	**2**	28.861546	0.637843	983
	3	29.105662	0.623862	989
	4	29.195469	0.637540	927

In [68]:
```
ctat_threshold = summary_week_hour['mean_avg_ctat'].quantile(0.75)
completion_threshold = summary_week_hour['mean_completion_rate'].quantile(0.25)
```

In [69]:
```
summary_week_hour['is_critical'] = (
    (summary_week_hour['mean_avg_ctat'] >= ctat_threshold) &
    (summary_week_hour['mean_completion_rate'] <= completion_threshold)
)
```

In [70]:
```
summary_week_hour[summary_week_hour['is_critical']==True]
```

Out[70]:

is_weekend	time_hour	mean_avg_ctat	mean_completion_rate	demand	is_critical
0	18	29.124183	0.610678	8841	True
1	6	29.178085	0.615186	1159	True
	14	29.163480	0.591424	2029	True

Ans. *Analysis shows that weekend Morning Rush (6 AM), Midday (2 PM) and weekday Evening Rush (6 PM) exhibit the highest average CTAT alongside the lowest completion rates, indicating severe supply–demand mismatch during these windows*

Reasons for low completion in

1. *Weekend Morning Rush : Late driver login, low perceived earnings vs effort*

2. *Weekend Midday : Driver break, drop location away from home, driver personal holiday*

3. *Weekday Evening Rush : Traffic and Driver selectivity--prefer short trips*

Targeted Interventions:

1. *Early login bonuses, Guaranteed minimum earnings for morning slots, time bound surge multipliers(6-8 AM)*

2. *Food coupons incentives, zone-specific incentives, weekend surge for midday slots*

3. *Long trip duration bonuses, congestion-adjusted incentives*

In [71]:

```
dataset.drop(columns=['is_weekend'], axis=1, inplace=True)
```

Q1 Only partial answer

Q2. **Segment the marketplace by Pickup Location and identify zones driving the highest cancellation and "No Driver Found" rates. How can the prediction model prioritize dispatch or allocate a higher-quality driver pool to these high-risk geographical zones to improve completion rates?**

In [72]:

```
data['is_completed']=(data['booking_status']=='Completed').astype(int)
data['is_no_driver_found']=(data['booking_status']=='No Driver Found').astype(int)
data['is_cancelled']=(data['booking_status'].isin(['Cancelled by Customer','Cancelled by Driver'])).astype(int)
```

In [73]:

```
pickup_summary = data.groupby('pickup_location').agg(
    total_requests=('booking_id', 'count'),
    completion_rate=('is_completed', 'mean'),
    no_driver_rate=('is_no_driver_found', 'mean'),
    cancelled_rate=('is_cancelled', 'mean'),
    mean_avg_ctat=('avg_ctat', 'mean')
).reset_index().round(2)
```

In [74]:

```
pickup_summary.sort_values(by='completion_rate', ascending=False)
```

Out[74]:

	pickup_location	total_requests	completion_rate	no_driver_rate	cancelled_rate	mean_avg_ctat
174	Welcome	836	0.66	0.05	0.24	29.08
156	Subhash Chowk	887	0.66	0.07	0.21	28.77
129	RK Puram	850	0.65	0.06	0.23	29.21
57	IMT Manesar	840	0.65	0.06	0.23	28.85
159	Sushant Lok	850	0.65	0.07	0.23	29.40
...	...	...	...	...	...	...
14	Badshahpur	868	0.58	0.08	0.27	29.09
124	Pragati Maidan	920	0.58	0.08	0.27	29.56
24	Chhatarpur	829	0.58	0.08	0.28	28.76
2	Akshardham	839	0.56	0.07	0.30	28.74
172	Vinobapuri	823	0.55	0.09	0.30	28.85

176 rows × 6 columns

In [75]:

```
request_threshold=pickup_summary['total_requests'].quantile(0.9)
no_driver_rate_threshold=pickup_summary['no_driver_rate'].quantile(0.8)
cancellation_rate_threshold=pickup_summary['cancelled_rate'].quantile(0.8)
```

In [76]:

```
high_risk_zone=pickup_summary[(pickup_summary['total_requests']>=request_threshold) & ((pickup_summary['no_driver_rate'
```

In [77]:

```
high_risk_zone
```

Out[77]:

	pickup_location	total_requests	completion_rate	no_driver_rate	cancelled_rate	mean_avg_ctat
0	AIIMS	918	0.61	0.07	0.26	29.16
42	Greater Kailash	895	0.62	0.07	0.26	29.29
96	Mehrauli	915	0.63	0.08	0.24	28.96
120	Pataudi Chowk	907	0.61	0.09	0.24	29.01
124	Pragati Maidan	920	0.58	0.08	0.27	29.56
141	Saket	931	0.60	0.07	0.27	28.89
151	Shivaji Park	900	0.62	0.05	0.27	29.02
165	Udyog Vihar	897	0.60	0.07	0.26	28.73
173	Vishwavidyalaya	895	0.62	0.05	0.26	28.89

Ans. ***The descriptive analysis using segregarion of pickup locations and aggregating on completion rate, cancellation rate and no driver found rates. Focusing on high-volume zones with structurally high failure rates, these 9 pickup locations are in high-risk zones.***

These zones must be mapped with high-reliability drivers, shorter acceptance windows and expanded incentives. A predictive modelling can be used downstream to operationalize this insight by prioritizing dispatch, expanding search-radius(without increasing acceptance window) and driver allocation in real time.

Q4. Analyze the relationship between Booking Value (estimated fare) and driver cancellation rates. Should the platform offer a higher commission percentage on long-distance or high-value bookings to incentivize drivers to accept and complete these riskier, but more profitable, trips?

In [78]:

```
data[data['booking_status']=='Cancelled by Driver']['booking_value'].unique()
```

Out[78]:

```
array([-1.])
```

Ans. ***In the dataset, booking value is not available to the driver-cancelled rides, which suggests drivers cancelled rides even before fare finalization. Therefore, we cannot establish direct relationship between booking_vlaue and driver cancellation from this data. This***

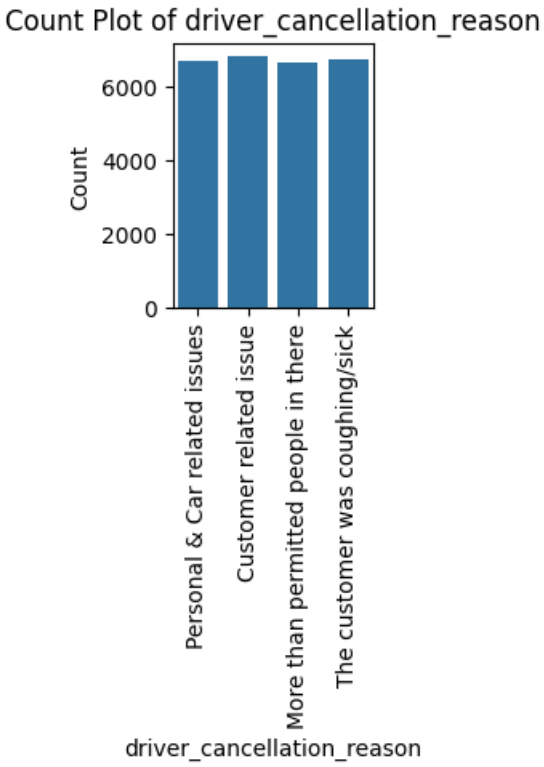
implies that increasing incentives would not prevent from driver cancellations. Instead, incentives should be designed around signals available at booking time-- pickup zone, estimated ctat, estimated vtat, or time of the day-- and post-acceptance bonuses can be used to encourage completion of longer or higher-valued trips.

Q5. Analyze the most frequent Driver Cancellation Reason (e.g., "Customer taking too long," "Not comfortable with drop"). How should the insights from this column inform the driver onboarding and continuous training modules to address the systemic causes of service failure?

In [79]:
```
data['driver_cancellation_reason'].unique()
```

Out[79]:
```
array([nan, 'Personal & Car related issues', 'Customer related issue',
       'More than permitted people in there',
       'The customer was coughing/sick'], dtype=object)
```

In [80]:
```
vis_cat_cols(data,['driver_cancellation_reason'], fig_size=(5,5))
```



In [81]:
```
driver_cancel_summary=data[data['booking_status']=='Cancelled by Driver'].groupby('driver_cancellation_reason').agg(tot
driver_cancel_summary['cancel_percent']=100*driver_cancel_summary['total_cancelled']/driver_cancel_summary['total_cance
driver_cancel_summary.sort_values(by='cancel_percent', ascending=False)
```

Out[81]:

driver_cancellation_reason	total_cancelled	cancel_percent
Customer related issue	6837	25.322222
The customer was coughing/sick	6751	25.003704
Personal & Car related issues	6726	24.911111
More than permitted people in there	6686	24.762963

Ans. *This analysis is primarily descriptive. On analysis of driver cancellations frequency distributions, since these are only populated for driver-cancelled rides. The most frequent reasons cluster around themes-- 'customer related issues', 'customer coughing or sick', 'personal & car related issues', 'more then permitted people in ride'.*

The reasons equally drive cancellations, highlighting systemic issues in customer's and driver's behavioral issues laced with safety concerns and pickup coordination.

These concerns directly inform requirement of driver training during onboarding, and setting clearer expectations around car health, clarify wait-time policies, communication with

customer, and acceptable cancellations.

At the same time, this analysis highlights customer-driven friction points, indicating a clearer customer-side expectation setting through in-app nudges, pickup readiness reminders, and education around driver safety considerations.

Addressing both these can help reduce driver-cancellations rather than treating them as separate incidents.

Q6. Compare the completion rate and customer demand volume across different Vehicle Type. Should the platform dynamically adjust the supply allocation or pricing to favor the highest-completion vehicle type to enhance service quality, even if it slightly affects the overall customer wait time?

In [82]:
```
vehicle_summary=data.groupby('vehicle_type').agg(total_demand=('booking_id','count'), completed=('is_completed','sum'))
vehicle_summary['completion_rate']=100*vehicle_summary['completed']/vehicle_summary['total_demand']
vehicle_summary['weighted_completion_rate']=np.log1p(vehicle_summary['total_demand'])*vehicle_summary['completion_rate'
```

In [83]:
```
vehicle_summary.sort_values(by='completion_rate', ascending=False)
```

Out[83]:

vehicle_type	total_demand	completed	completion_rate	weighted_completion_rate
Uber XL	4449	2783	62.553383	5.254897
Bike	22517	14034	62.326242	6.246380
Go Mini	29806	18549	62.232436	6.411496
Premier Sedan	18111	11252	62.127989	6.091233
eBike	10557	6551	62.053614	5.749043
Auto	37419	23155	61.880328	6.515974
Go Sedan	27141	16676	61.442099	6.272524

In [84]:
```
vehicle_summary.sort_values(by='weighted_completion_rate', ascending=False)
```

Out[84]:

vehicle_type	total_demand	completed	completion_rate	weighted_completion_rate
Auto	37419	23155	61.880328	6.515974
Go Mini	29806	18549	62.232436	6.411496
Go Sedan	27141	16676	61.442099	6.272524
Bike	22517	14034	62.326242	6.246380
Premier Sedan	18111	11252	62.127989	6.091233
eBike	10557	6551	62.053614	5.749043
Uber XL	4449	2783	62.553383	5.254897

Ans. *While there are small differences in completion rate across vehicle type, they are tightly clustered around ~62%, indicating that the vehicle type alone is not a strong indicator of completion. Higher weighted scores for Auto primarily reflect its dominant market share, not superior reliability. Therefore, globally favoring one vehicle over the other would not materially improve service quality and can hurt.*

Instead, supply allocation and pricing should be driven by contextual factors like zone-level risk, driver reliability, time based demand. Vehicle_type becomes a secondary constraint than a primary lever. For example, bikes during peak congestions in morning rush.

Data Modelling

In [85]:
```
dataset.isna().sum()
```

Out[85]:

	0
booking_status_bin	0
vehicle_type	0
pickup_location	0
drop_location	0
avg_vtat	0
avg_ctat	0
time_hour	0
time_slot_of_day	0
day_type	0
pickup_location_density_score	0
drop_location_density_score	0

dtype: int64

Dataset is without nulls now

In [86]:

```
cat_cols, num_cols =find_cat_num_cols(dataset)
cat_cols, num_cols
```

Out[86]:

```
(['booking_status_bin', 'vehicle_type', 'time_slot_of_day', 'day_type'],
 ['pickup_location',
  'drop_location',
  'avg_vtat',
  'avg_ctat',
  'time_hour',
  'pickup_location_density_score',
  'drop_location_density_score'])
```

Code for Modelling

In [87]:

```
def remove_outliers(df, num_cols):
  z_scores=stats.zscore(df[num_cols])
  outliers=np.abs(z_scores)>1.5
  filtered_rows=np.any(outliers,axis=1)
  cleaned_data=df[~filtered_rows].reset_index(drop=True)
  return cleaned_data
```

In [88]:

```
def scale_data(cleaned_data, num_columns, cat_columns):
  num_data=cleaned_data[num_columns]
  cat_data=cleaned_data[cat_columns]
  scaler=StandardScaler()
  scaled_num_data=pd.DataFrame(scaler.fit_transform(num_data),columns=num_columns)
  scaled_data = pd.concat([scaled_num_data,cat_data],axis=1)
  return scaled_data

# clean_data=remove_outliers(data,num_cols)
# scaled_data = scale_data(clean_data,num_cols,cat_cols)
# scaled_data
```

In [145…

```
def run_model_without_smote(df, model, name):
  x=df.drop('booking_status_bin', axis=1)
  y=df['booking_status_bin'].astype(int) # Explicitly cast y to int here
  x_train, x_test, y_train, y_test = train_test_split(x,y,stratify=y, test_size=0.2, random_state=42)

  model.fit(x_train,y_train)
  y_pred=model.predict(x_test)

  return y_pred,y_test, x_test.index, model, x_test, y_test
```

In [146…

```
from sklearn.metrics import f1_score
def models_no_smote(df, fairness_df):
    results = []  # Initialize a list to store results
    cm_data = []  # Initialize a list to store confusion matrix data
    roc_data = []  # Initialize a list to store ROC curve data
```

```python
fairness_data = []
models = {
    "Logistic Regression": LogisticRegression(
        C=0.1,
        class_weight='balanced',
        max_iter=1000
    ),

    "Gradient Boosting": GradientBoostingClassifier(
        n_estimators=200,
        learning_rate=0.05,
        max_depth=3
    ),

    "AdaBoost": AdaBoostClassifier(
        n_estimators=200,
        learning_rate=0.05
    ),

    "Gaussian Naive Bayes": GaussianNB(),

    "K-Nearest Neighbors": KNeighborsClassifier(
        n_neighbors=15,
        weights='distance'
    ),

    "Decision Tree": DecisionTreeClassifier(
        max_depth=5,
        min_samples_leaf=100,
        class_weight='balanced'
    ),

    "Random Forest": RandomForestClassifier(
        n_estimators=300,
        max_depth=6,
        min_samples_leaf=100,
        class_weight='balanced',
        random_state=42
    )
}

# Iterate through the models
for name, model in models.items():
    # Run the model and get predictions and actual labels
    predictions, y_test, test_idx, model_after_fit, x_val, y_val = run_model_without_smote(df, model, name)

    # Evaluate the model and calculate metrics
    accuracy, precision, recall, f1, cm = evaluate_model(y_test, predictions)

    # Calculate ROC curve data
    fpr, tpr, _ = roc_curve(y_test, predictions)
    roc_auc = auc(fpr, tpr)

    # Store the results in a dictionary
    results.append({
        "Model": name,
        "Accuracy": accuracy,
        "Precision": precision,
        "Recall": recall,
        "F1 Score": f1,
        "Model after fit": model_after_fit,
        "X Val":x_val,
        "Y Val":y_val
    })
    test_df = fairness_df.loc[test_idx].copy()
    print(test_df.columns)
    test_df['y_true'] = y_test.values
    test_df['y_pred'] = predictions

    for bucket, bucket_df in test_df.groupby('pickup_zone'):
        if len(bucket_df) < 50:  # stability guard
            continue
        bucket_f1 = f1_score(bucket_df['y_true'], bucket_df['y_pred'])

        fairness_data.append({
            "Model": name,
            "pickup_zone": bucket,
            "F1 Score": round(bucket_f1, 4),
            "Support": len(bucket_df)
        })

    cm_data.append({
```

```
            "Model": name,
            "Matrix": cm
        })

        # Store ROC curve data in a dictionary
        roc_data.append({
            "Model": name,
            "FPR": fpr.tolist(),
            "TPR": tpr.tolist(),
            "AUC": roc_auc
        })

    # Create a DataFrame from the results, round the values, and sort by accuracy
    results_df = pd.DataFrame(results).round(4).sort_values(by="Accuracy", ascending=False)
    fairness_df = pd.DataFrame(fairness_data).round(4).sort_values(by="F1 Score", ascending=False)

    return results_df, cm_data, roc_data, fairness_df
```

In [120…

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix

def evaluate_model(y_true, y_pred):
    # Calculate accuracy
    accuracy = accuracy_score(y_true, y_pred)

    # Calculate precision
    precision = precision_score(y_true, y_pred)

    # Calculate recall
    recall = recall_score(y_true, y_pred)

    # Calculate F1 score
    f1 = f1_score(y_true, y_pred)

    # Calculate the confusion matrix
    cm = confusion_matrix(y_true, y_pred)

    return accuracy, precision, recall, f1, cm
```

In [147…

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler

def pipeline_no_smote(df):
    fairness_cols = ['pickup_zone']
    fairness_df = df[fairness_cols].copy()

    y = df['booking_status_bin'].copy()
    X = df.drop(columns=['booking_status_bin'] + fairness_cols, axis=1).copy()

    categorical_cols = X.select_dtypes(include=['object', 'category']).columns.tolist()
    numerical_cols = X.select_dtypes(include=['int64', 'float64']).columns.tolist()

    # numerical_features_for_processing = X.columns.tolist()

    combined_data_for_outliers = pd.concat([X, y.rename('booking_status_bin')], axis=1)

    cleaned_combined_data = remove_outliers(combined_data_for_outliers, numerical_cols)

    y_processed = cleaned_combined_data['booking_status_bin'].reset_index(drop=True)
    X_processed = cleaned_combined_data.drop('booking_status_bin', axis=1).reset_index(drop=True)

    fairness_processed = fairness_df.loc[X_processed.index].reset_index(drop=True)

    preprocessor = ColumnTransformer(
        transformers=[
            ('num', StandardScaler(), numerical_cols),
            ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False), categorical_cols)
        ],
        remainder='drop'
    )

    X_transformed = preprocessor.fit_transform(X_processed)

    ohe_feature_names = preprocessor.named_transformers_['cat'].get_feature_names_out(categorical_cols)
    final_feature_names = numerical_cols + list(ohe_feature_names)

    X_final = pd.DataFrame(
        X_transformed,
        columns=final_feature_names
    )
```

```
    final_df = pd.concat([X_final, y_processed], axis=1)

    performance, cm_data, roc_data, fairness = models_no_smote(final_df, fairness_processed)
    return performance, cm_data, roc_data, fairness
    # # Since all features in `X_processed` are intended for scaling, `cat_cols` will be an empty list.
    # scaled_X = scale_data(X_processed, numerical_cols, [])

    # # It's crucial that `y_processed` has its index reset to align correctly with `scaled_X`.
    # final_data_for_models = pd.concat([scaled_X, y_processed.reset_index(drop=True)], axis=1)

    # performance, cm_data, roc_data = models_no_smote(final_data_for_models)

    # return performance, cm_data, roc_data
```

In [93]:

```
def plot_confusion_matrix(cm, labels, model_name, ax):
    # Create an image of the confusion matrix
    im = ax.imshow(cm, interpolation='nearest')

    # Add a colorbar to the plot
    ax.figure.colorbar(im, ax=ax)

    # Set axis labels and title
    ax.set(xticks=np.arange(cm.shape[1]),
           yticks=np.arange(cm.shape[0]),
           xticklabels=labels, yticklabels=labels,
           title=f'Confusion Matrix for {model_name}',
           ylabel='True label',
           xlabel='Predicted label')

    # Add text annotations for each cell in the matrix
    for i in range(cm.shape[0]):
        for j in range(cm.shape[1]):
            ax.text(j, i, str(cm[i, j]), ha="center", va="center", color="w")
```

In [94]:

```
def plot_confusion_matrices(cm_data, columns, rows):
    # Determine the number of models and create subplots accordingly
    num_models = len(cm_data)

    # Calculate the actual number of rows needed based on num_models and desired columns
    actual_rows = (num_models + columns - 1) // columns

    fig, axes = plt.subplots(actual_rows, columns, figsize=(15, 10))

    # Flatten the axes array to easily iterate through each subplot
    # This handles cases where actual_rows or columns might be 1, resulting in a 1D axes array
    axes = axes.flatten()

    # Iterate through the confusion matrix data for each model
    for i, entry in enumerate(cm_data):
        model_name = entry['Model']
        cm = entry['Matrix']
        class_labels = ['Incomplete', 'Completed']

        plot_confusion_matrix(cm, class_labels, model_name, axes[i])

    # Turn off any empty subplots if the grid has more slots than models
    for i in range(num_models, actual_rows * columns):
        axes[i].axis('off')

    # Adjust the layout and display the subplots
    plt.tight_layout()
    plt.show()
```

In [114…

```
zone_labels=['low', 'medium','high']
p33=dataset['pickup_location_density_score'].quantile(0.33)
p66=dataset['pickup_location_density_score'].quantile(0.66)
dataset['pickup_zone']=pd.cut(dataset['pickup_location_density_score'],bins=[0,p33,p66,np.inf],labels=zone_labels,right=
dataset['pickup_zone'].value_counts()
```

Out[114…

pickup_zone	count
high	51657
medium	49410
low	48933

dtype: int64

In [116…

```
dataset.drop(columns=['avg_vtat','avg_ctat'], axis=1, inplace=True, errors='ignore')
dataset.columns
```

Out[116…

```
Index(['booking_status_bin', 'vehicle_type', 'pickup_location',
       'drop_location', 'time_hour', 'time_slot_of_day', 'day_type',
       'pickup_location_density_score', 'drop_location_density_score',
       'pickup_zone'],
      dtype='object')
```

In [178…

```
import time
start_time = time.perf_counter()
# pipeline_no_smote returns (performance_df, cm_data, roc_data, fairness_df)
full_model_results, cm_data_no_smote, roc_data_no_smote, fairness = pipeline_no_smote(dataset)
end_time = time.perf_counter()
time_taken_no_smote=np.round(end_time - start_time,2)

# Store the full results for SHAP analysis
models_after_fit = full_model_results.copy()

# Prepare the performance dataframe for display by dropping internal columns
performance_no_smote = full_model_results.drop(columns=['Model after fit', 'X Val', 'Y Val'], axis=1, errors='ignore')
```

```
Index(['pickup_zone'], dtype='object')
/tmp/ipython-input-3963442004.py:75: FutureWarning: The default of observed=False is deprecated and will be changed to Tru
e in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future defau
lt and silence this warning.
  for bucket, bucket_df in test_df.groupby('pickup_zone'):
Index(['pickup_zone'], dtype='object')
/tmp/ipython-input-3963442004.py:75: FutureWarning: The default of observed=False is deprecated and will be changed to Tru
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Index(['pickup_zone'], dtype='object')
/tmp/ipython-input-3963442004.py:75: FutureWarning: The default of observed=False is deprecated and will be changed to Tru
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Index(['pickup_zone'], dtype='object')
/tmp/ipython-input-3963442004.py:75: FutureWarning: The default of observed=False is deprecated and will be changed to Tru
e in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future defau
lt and silence this warning.
  for bucket, bucket_df in test_df.groupby('pickup_zone'):
```

In [179…

```
print(f"Time taken to run models: {time_taken_no_smote} secs")

# 'performance_no_smote' is now prepared for display in the previous cell.
performance_no_smote.sort_values(by='F1 Score',ascending=False).round(2)
```

```
Time taken to run models: 254.0 secs
```

Out[179…

	Model	Accuracy	Precision	Recall	F1 Score
2	AdaBoost	0.62	0.62	1.00	0.77
1	Gradient Boosting	0.62	0.62	1.00	0.77
4	K-Nearest Neighbors	0.58	0.62	0.82	0.71
5	Decision Tree	0.54	0.63	0.64	0.63
6	Random Forest	0.51	0.63	0.53	0.58
3	Gaussian Naive Bayes	0.50	0.61	0.52	0.56
0	Logistic Regression	0.49	0.61	0.50	0.55

In [128…

```
fairness.sort_values(by='F1 Score',ascending=False).round(4)
```

Out[128…

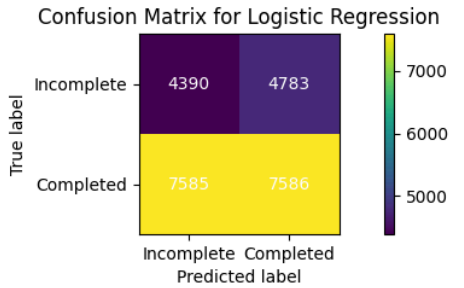
	Model	pickup_zone	F1 Score	Support
7	AdaBoost	medium	0.7716	8024
4	Gradient Boosting	medium	0.7710	8024
6	AdaBoost	low	0.7699	7893
3	Gradient Boosting	low	0.7695	7893
8	AdaBoost	high	0.7624	8427
5	Gradient Boosting	high	0.7616	8427
12	K-Nearest Neighbors	low	0.7133	7893
13	K-Nearest Neighbors	medium	0.7101	8024
14	K-Nearest Neighbors	high	0.6994	8427
16	Decision Tree	medium	0.6357	8024
15	Decision Tree	low	0.6332	7893
17	Decision Tree	high	0.6323	8427
19	Random Forest	medium	0.5845	8024
20	Random Forest	high	0.5737	8427
18	Random Forest	low	0.5727	7893
10	Gaussian Naive Bayes	medium	0.5705	8024
11	Gaussian Naive Bayes	high	0.5624	8427
9	Gaussian Naive Bayes	low	0.5614	7893
1	Logistic Regression	medium	0.5539	8024
2	Logistic Regression	high	0.5499	8427
0	Logistic Regression	low	0.5489	7893

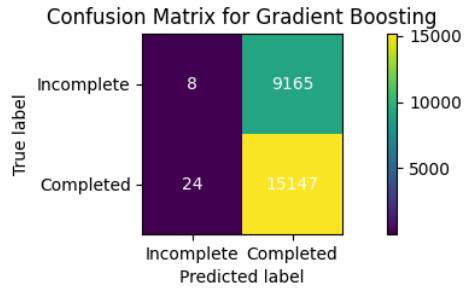
Q7. Evaluate the model's prediction fairness by comparing its F1 -score across different major pickup zones (e.g., North vs. South NCR). If performance disparities are found, what mitigation strategy (e.g., collecting more data from minority zones, using balanced loss) must be implemented to prevent service inequality?

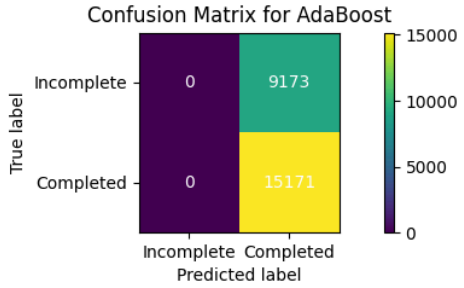
Ans. *To evaluate fairness, I compared F1 scores across pickup locations segmented into low, medium, high based on pickup_location_density_score that is based on historical demand. The results show highly consistent performance across all density groups for every model, with minor variations. Medium density zone exhibit slightly higher F1 scores, while high-density zones exhibit marginally lower, likely due to complex and competitive supply-demand dynamics rather than model bias. Since no group is systematically disadvantaged, no fairness related mitigation such as reweighting or oversapling is required. Improvements should instead focus on enriching real-time operational features, particulrly in high-density zones.*

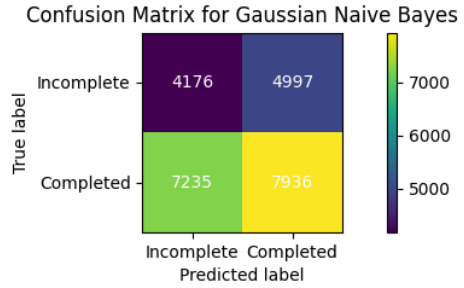
In [127…

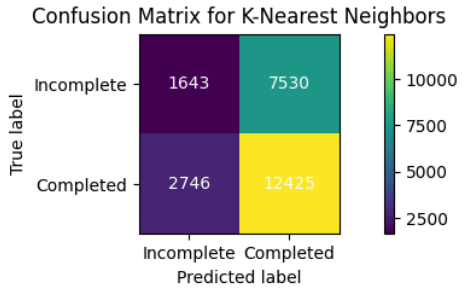
```
plot_confusion_matrices(cm_data_no_smote,columns=2,rows=3)
```

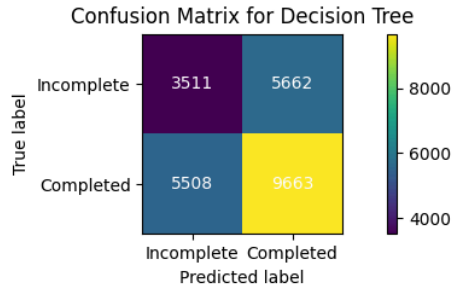


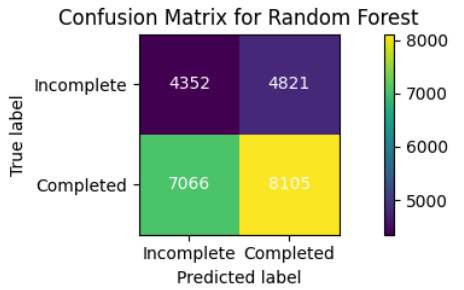












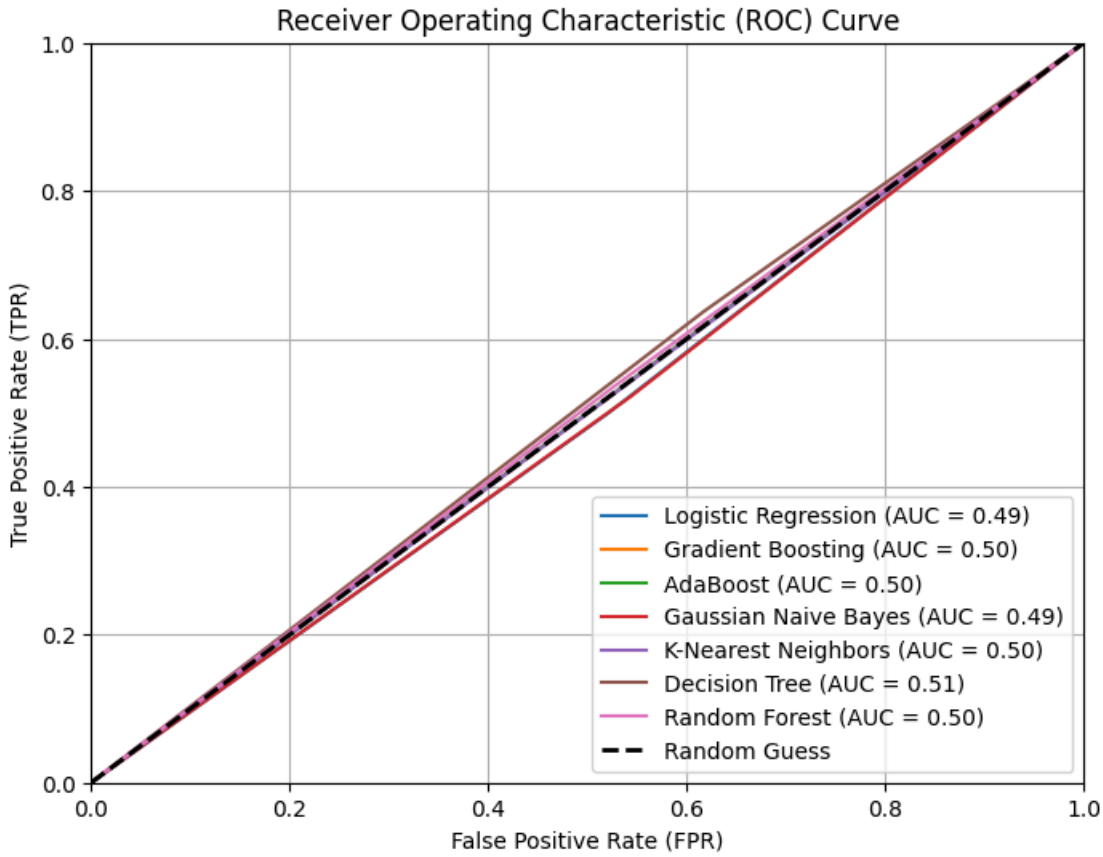
The confusion matrices indicate that models either collapse into predicting majority class (AdaBoost, GradientBoost) or behave close to random, missing a large fraction of incomplete rides (Random Forest, Decision Tree, Logistic Regression, Gaussian Naive) or slightly miss failures but still weak (KNN). This confirms that after removing leakage columns, request-time features lack sufficient discriminatory power. Even ensemble models cannot meaningfully separate failures from successful rides, indicating that successfull completion is driven by real-time supply-demand dynamics than static booking attributes.

In [99]:

```
def plot_roc_curve(roc_data):
    plt.figure(figsize=(8, 6))
    plt.title('Receiver Operating Characteristic (ROC) Curve')

    for model_data in roc_data:
        model_name = model_data['Model']
        fpr = model_data['FPR']
        tpr = model_data['TPR']
        auc = model_data['AUC']

        plt.plot(fpr, tpr, label=f'{model_name} (AUC = {auc:.2f})')

    plt.plot([0, 1], [0, 1], 'k--', lw=2, label='Random Guess')
    plt.xlim([0.0, 1.0])
    plt.ylim([0.0, 1.0])
    plt.xlabel('False Positive Rate (FPR)')
    plt.ylabel('True Positive Rate (TPR)')
    plt.legend(loc='lower right')
    plt.grid(True)
    plt.show()
```

In [100…

```
plot_roc_curve(roc_data_no_smote)
```



After removing post-outcome leakage columns, and tuning conservatively, all models converge to near baseline performance (auc=0.5). Boosting models achieve high f1_scrore by predicting majority class, which is operationally not useful. This implies that ride completion is driven by real-time supply dynamics not present in the dataset, making this suitable for rule-based or decision-support systems rather than pure-prediction.

Q3. **Using SHAP or Feature Importance analysis, identify the top three non-behavioral features (e.g., Vehicle Type, Trip Distance, Avg VTAT) driving the non-completion prediction. Which operational levers (e.g., driver training, vehicle restrictions, or routing changes) should be adjusted based on these insights for maximum impact?**

In [150…

```
models_after_fit
```

Out[150…

	Model	Accuracy	Precision	Recall	F1 Score	Model after fit
2	AdaBoost	0.6232	0.6232	1.0000	0.7679	(DecisionTreeClassifier(max_depth=1, random_st...
1	Gradient Boosting	0.6225	0.6230	0.9984	0.7673	([DecisionTreeRegressor(criterion='friedman_ms...
4	K-Nearest Neighbors	0.5779	0.6227	0.8190	0.7075	KNeighborsClassifier(n_neighbors=15, weights='...
5	Decision Tree	0.5412	0.6305	0.6369	0.6337	DecisionTreeClassifier(class_weight='balanced'...
6	Random Forest	0.5117	0.6270	0.5342	0.5769	(DecisionTreeClassifier(max_depth=6, max_featu...
3	Gaussian Naive Bayes	0.4975	0.6136	0.5231	0.5648	GaussianNB()
0	Logistic Regression	0.4919	0.6133	0.5000	0.5509	LogisticRegression(C=0.1, class_weight='balanc...

Though KNN showed slightly better balance in classification, it lacks interpretability for SHAP(values are unstable, conceptually weak). Therefore, Decision tree is used for SHAP analysis to extract stable and feature dependence explanations.

In [182…

```
DT_model=models_after_fit[models_after_fit['Model']=='Decision Tree']['Model after fit'].iloc[0]
```

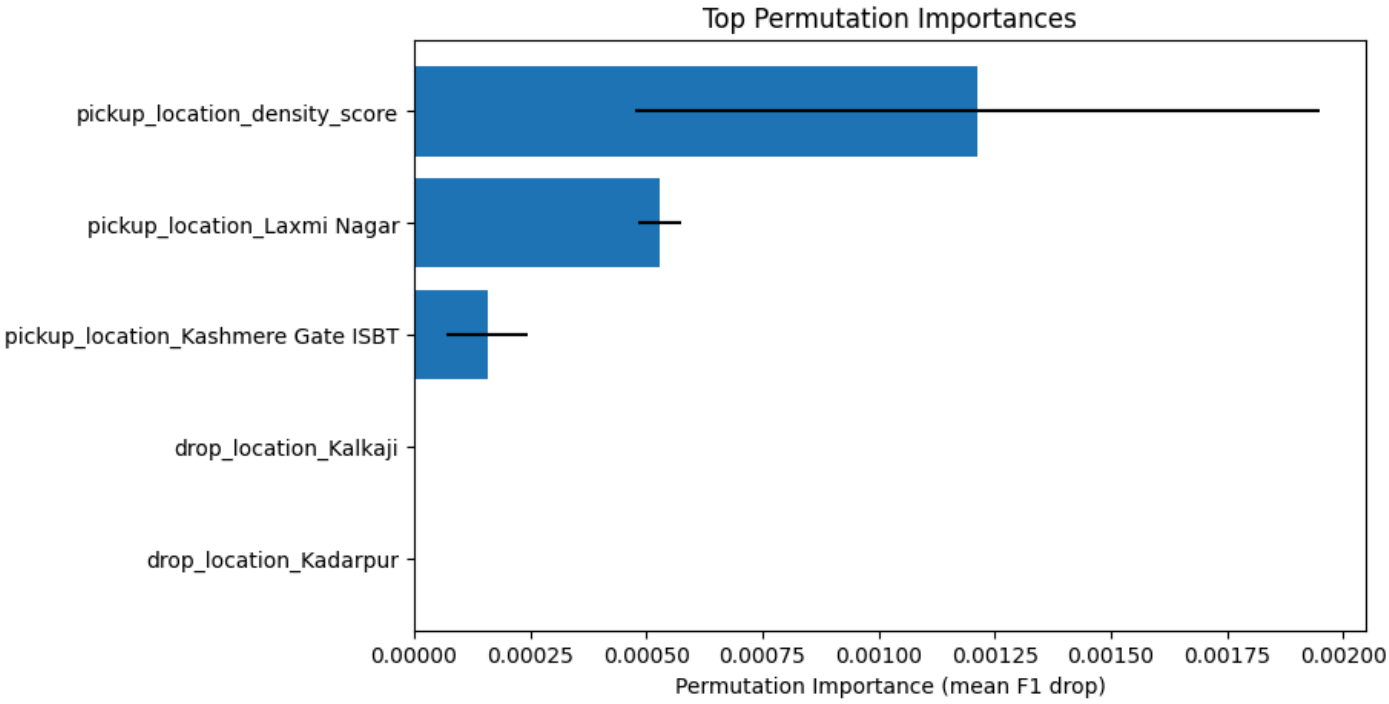
In [ ]:

```
dt_row_index = performance_no_smote[performance_no_smote['Model'] == 'Decision Tree'].index[0]

x_val_dt_model = models_after_fit.loc[dt_row_index, 'X Val']
y_val_dt_model = models_after_fit.loc[dt_row_index, 'Y Val']
```

In [189…

```
# permutation importance = if I destroy this feature how it is going to effect F1 score
```

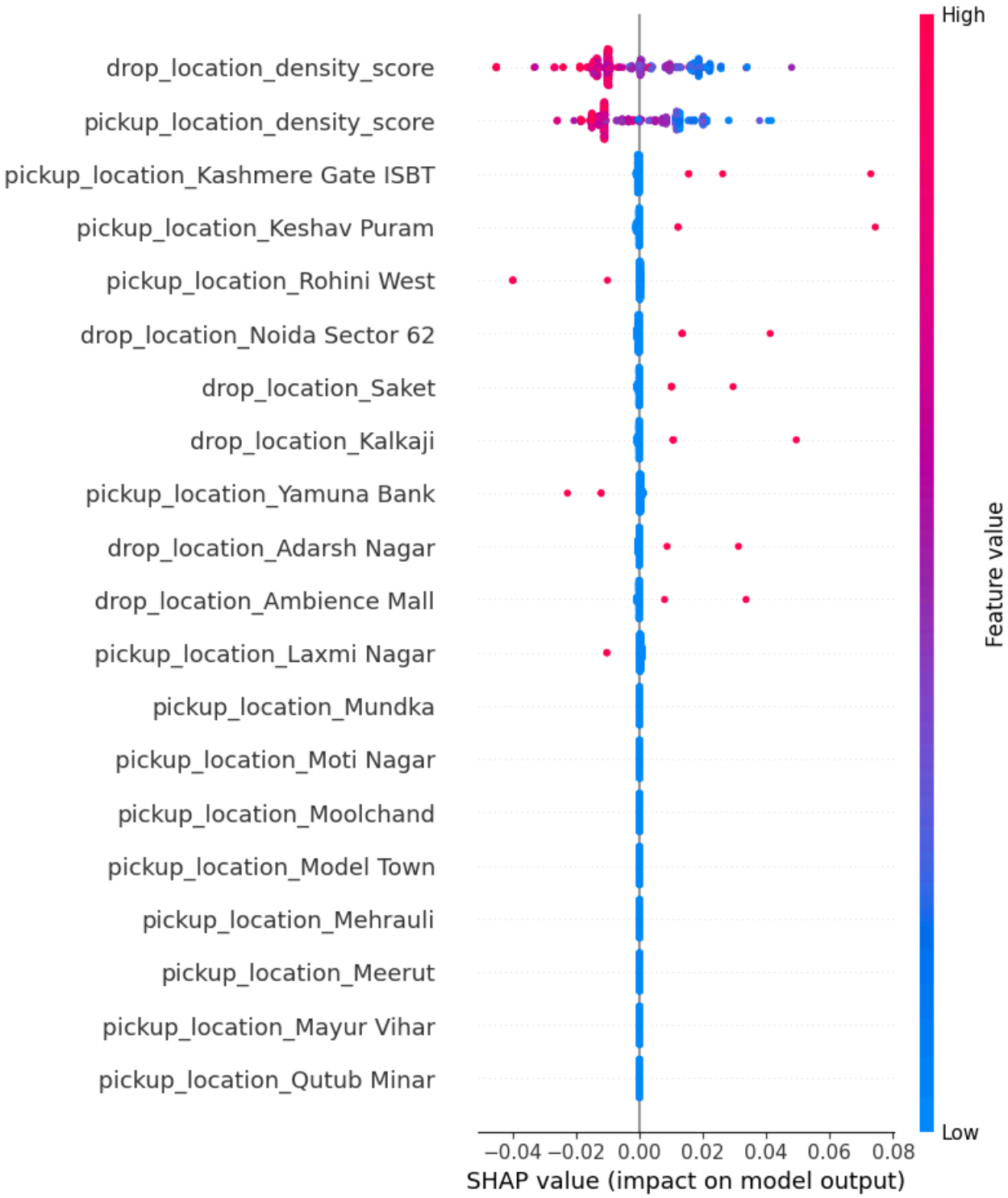
```
from sklearn.inspection import permutation_importance

# Use x_val_dt_model and y_val_dt_model for permutation importance
perm = permutation_importance(
    DT_model,
    x_val_dt_model,
    y_val_dt_model,
    scoring='f1',
    n_repeats=5,
    random_state=42
)
imp = pd.DataFrame({
    "feature": x_val_dt_model.columns,
    "importance_mean": perm.importances_mean,
    "importance_std": perm.importances_std
}).sort_values("importance_mean", ascending=False)

plt.figure(figsize=(8, 5))
plt.barh(imp["feature"][:5], imp["importance_mean"][:5], xerr=imp["importance_std"][:5])
plt.gca().invert_yaxis()
plt.xlabel("Permutation Importance (mean F1 drop)")
plt.title("Top Permutation Importances")
plt.show()
```



The marginal F1-score reductions observed for pickup_location_density_score and certain pickup locations (e.g., Laxmi Nagar, Kashmere Gate ISBT) are statistically insignificant (~0.00125), suggesting minimal sensitivity to their exclusion. Therefore, SHAP values were used as the primary basis for feature importance interpretation

In [187…

```
#SHAP for feature importance
import shap

X_sample = x_val_dt_model.sample(n=500, random_state=42)

explainer = shap.TreeExplainer(DT_model)
shap_values = explainer.shap_values(X_sample)

# For a binary classification model where shap_values is a 3D array (n_samples, n_features, n_classes),
shap_values_for_class_0 = shap_values[:, :, 0]

# Ensure consistency before plotting
assert DT_model.n_features_in_ == X_sample.shape[1], "Model and X_sample feature count mismatch!"
assert shap_values_for_class_0.shape == (X_sample.shape[0], X_sample.shape[1]), "Shape of extracted SHAP values for cla

shap.summary_plot(shap_values_for_class_0, X_sample)
```



Ans. *SHAP Analysis on interpretable Decision Tree model shows us that non-completion risk is primarily influenced by loaction based and non-behavioral features.*

Only small set of features influence non-completion.

Top drivers of non-completion(Non-behavioral):

1. *drop_location_density_score: Higher density tend to have lower probability of non-completion, can be due to driver availability or infrastructure availability or more driver familiarity. Similarly, low density scores have higher chaces of non-completion due to poor navigation or peripheral location or uncertainity of return demand or safety concerns of driver.*

2. *pickup_location_density_score: Similarly, high reluctance to acceptance for low density score zones--poor navigation metrics, poor communication with customer, safety concerns, etc. High acceptance to high density zone due to driver awareness on location, proximity to commercial hubs, better road access, etc.*

3. *Other features(mostly ohe pickup locations) have dots mostly centered around 0 with ocassional red dots on positive side of SHAP indicating these locations--like*

Kashmere_Gate_ISBT, Keshav_Puram-- have likely chances of non-completion when they are pickup locations.

Therefore, maximum impact will come from operational levers like driver positioning, zone specific incentives, and improving routing support in less dense drop locations rather than model-based decisioning.